1.1 Sum of Two Digits

Sum of Two Digits Problem
Compute the sum of two single digit numbers.

Input: Two single digit numbers.
Output: The sum of these numbers.

$$2 + 3 = 5$$

We start from this ridiculously simple problem to show you the pipeline of reading the problem statement, designing an algorithm, implementing it, testing and debugging your program, and submitting it to the grading system.

Input format. Integers a and b on the same line (separated by a space).

Output format. The sum of a and b.

Constraints. $0 \le a, b \le 9$.

Sample.

Input:

```
9 7
```

Output:

```
16
```

Time limits (sec.):

C++	Java	Python	C	C#	Go	Haskell	JavaScript	Kotlin	Ruby	Rust	Scala
1	1.5	5	1	1.5	1.5	2	5	1.5	5	1	3

Memory limit. 512 Mb.

1.1.1 Solutions in Various Programming Languages

For this trivial problem, we will skip "Designing an algorithm" step and will move right to the pseudocode.

```
SumOfTwoDigits(a, b):
return a + b
```

Since the pseudocode does not specify how we input a and b, below we provide solutions in C++, Java, and Python3 programming languages as well as recommendations on compiling and running them. You can copy-and-paste the code to a file, compile/run it, test it on a few datasets, and then submit (the source file, not the compiled executable) to the grading system. Needless to say, we assume that you know the basics of one of programming languages that we use in our grading system.

C++

```cpp
#include <iostream>

int sum_of_digits(int first, int second) {
    return first + second;
}

int main() {
    int a = 0;
    int b = 0;
    std::cin >> a;
    std::cin >> b;
    std::cout << sum_of_digits(a, b);
    return 0;
}
```

Save this to a file (say, `aplusb.cpp`), compile it, run the resulting executable, and enter two numbers (on the same line).

Java

```java
import java.util.Scanner;

class SumOfDigits {
    static int sumOfDigits(int first_digit,
        int second_digit) {
        return first_digit + second_digit;
    }

    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);
        int a = s.nextInt();
        int b = s.nextInt();
        System.out.println(sumOfDigits(a, b));
    }
}
```

Save this to a file `SumOfDigits.java`, compile it, run the resulting executable, and enter two numbers (on the same line).

Python

```python
def sum_of_digits(first_digit, second_digit):
    return first_digit + second_digit

if __name__ == '__main__':
    a, b = map(int, input().split())
    print(sum_of_digits(a, b))
```

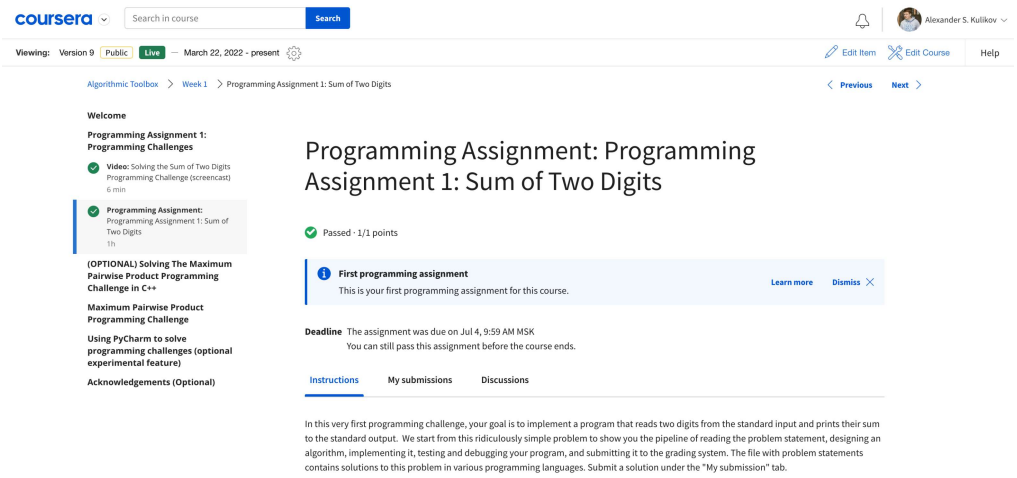
Save this to a file (say, `aplusb.py`), run it, and enter two numbers on the same line.

Your goal is to implement an algorithm that produces a correct result under the given time and memory limits for any input satisfying the given constraints. You do not need to check that the input data satisfies the constraints, e.g., for the Sum of Two Digits Problem you do not need to

check that the given integers a and b are indeed single digit integers (this is guaranteed).

1.1.2 Submitting to the Grading System at Coursera

This is what a fresh submission page looks like:



If this is you first programming assignment at Coursera, we encourage you to go through general information about programming assignments ("Learn more" button in the blue bubble).

The line below the blue bubble indicates the deadline for this assignment. Please not that this is a suggestion rather than a hard deadline. If you miss a deadline, you may safely switch to the next session of the course (a new session starts automatically every two weeks). All your progress will be transferred to the new session in this case. See Coursera help article on switching sessions.

The "Discussion" tab leads to a forum attached to this assignment. Please post your questions and help other learners there.

To submit your solution, go to the "My submissions" tab, press the "Create submission" button, upload the source code of your solution, and press the "Submit" button. You may safely leave the page while your solution is being graded (in most cases, it is done in less than a minute, but when the servers are overloaded it may take several minutes; please be patient).